A *GitHub Action* to check your code with *diKTat*

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[GitHub release] | *https://badgen.net/github/release/saveourtool/benedikt/latest?color=green*

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> An always updated version of this document is available here as a PDF e-book.

Features

- Customizable `diktat-analysis.yml` location. You can use the rule set configuration with an alternate name or at a non-default location.
- Customizable JVM vendor and version. You can run *diKTat* using a default JVM, or you can set up your own one.
- Customizable reporters (*SARIF*, *JSON*, *Checkstyle* XML).
- Allows multiple input paths. If you have a multi-module project and only wish to check certain

directories or modules, you can configure the action accordingly.

- The last line of the log output reported to the summary page:

Usage

In the simplest scenario, the action can be used without input parameters; you just need to check out your code first, using `actions/checkout`:

```
jobs:
  diktat_check:
    name: 'diKTat Check'
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3

      - uses: saveourtool/benedikt@v1
```

Configuration

`config`: custom configuration file

- Default: `diktat-analysis.yml`
- Required: **no**

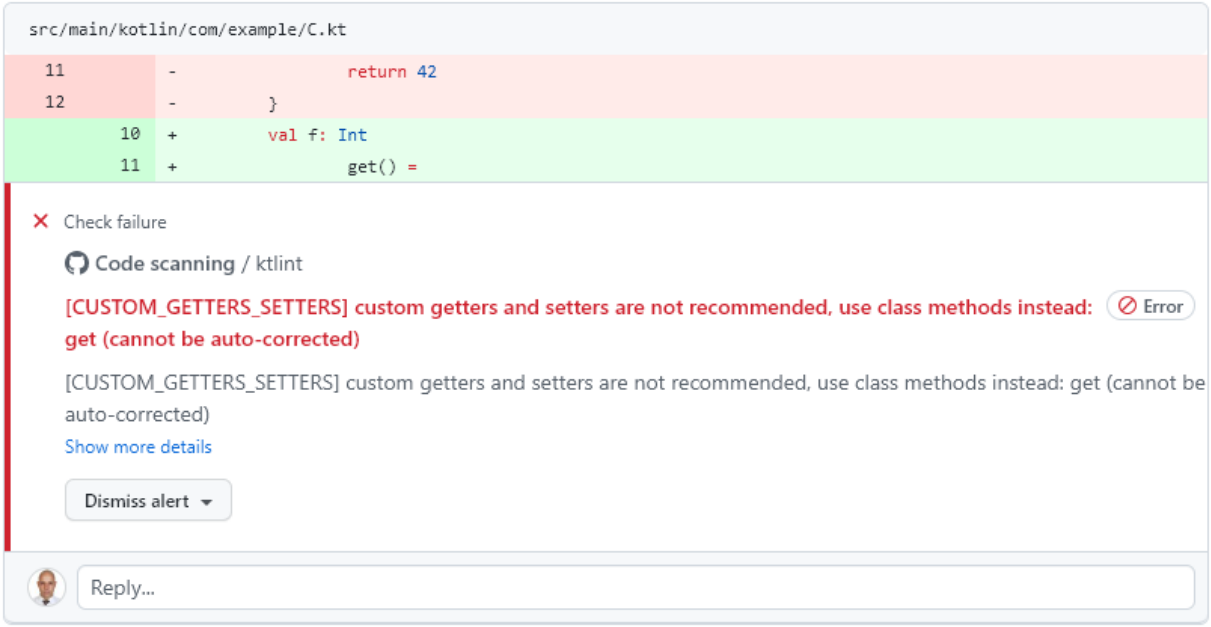
You can override the name or the path of your YAML configuration file using the `config` input parameter, e. g.:

```
      - uses: saveourtool/benedikt@v1
        with:
          config: path/to/diktat-analysis-custom.yml
```

`reporter`: requesting a custom reporter

If you wish, you can report errors in a custom format.

- Default: `sarif`
- Required: **no**
- Possible values: any of the following.
  - `sarif`: report errors in the *SARIF* format. The output file will be named `report.sarif` and automatically uploaded to *GitHub* using the `upload-sarif` action. This will enable the check results to be shown as annotations in the pull request:



  as well as in the *Code scanning alerts* section of your repository:

Code scanning alerts / #396

[MAGIC_NUMBER] avoid using magic numbers, instead define constants with clear names describing what the magic number means: 42 (cannot be auto-corrected)

Dismiss alert

Open in master 8 minutes ago

src/main/kotlin/com/example/C.kt:19

```
16          */
17         val g: Int
18                 get() =
19                         42
```

[MAGIC_NUMBER] avoid using magic numbers, instead define constants with clear names describing what the magic number means: 42 (cannot be auto-corrected)

ktlint

```
20  }
```

Tool	Rule ID
ktlint	diktat-ruleset:magic-number

No rule help available for this alert.

Severity

Error

Affected branches

master

First detected in commit 8 minutes ago

Initiate a new code scanning procedure 5511f17

src/main/kotlin/com/example/C.kt:19 on branch master

Appeared in branch master 8 minutes ago

diKTat Check, Ubuntu Linux #62: Commit 5511f177

- checkstyle: this is the reporter of choice if you ever encounter issues with the sarif reporter. Errors are reported in the *Checkstyle-XML* format to the file named checkstyle-report.xml. The report is then consumed by the reviewdog tool and uploaded to *GitHub*, resulting in code annotations similar to those produced by the sarif reporter:

```
2
3  + /**
4  +  * This is a simple class covered by both unit and integration tests.
5  +  */
6    object C {
```

Check failure on line 6 in src/main/kotlin/com/example/C.kt

GitHub Actions / diKTat errors reported by reviewdog

[diKTat errors reported by reviewdog] src/main/kotlin/com/example/C.kt#L6 <diktat-ruleset:identifier-naming>

[OBJECT_NAME_INCORRECT] object structure name should be in PascalCase and should contain only latin (ASCII) letters or numbers

Raw output

- plain: report errors in the plain-text format, e. g.:

```
C.kt:1:1: [MISSING_KDOC_TOP_LEVEL] all public and internal top-level classes and
functions should have Kdoc: C (cannot be auto-corrected) (diktat-ruleset:kdoc-
comments)
C.kt:1:1: [FILE_NAME_INCORRECT] file name is incorrect - it should end with .kt
extension and be in PascalCase: C.kt (diktat-ruleset:file-naming)
C.kt:1:1: [PACKAGE_NAME_MISSING] no package name declared in a file: C.kt
```

```
(diktat-ruleset:package-naming)
C.kt:1:7: [CLASS_NAME_INCORRECT] class/enum/interface name should be in
PascalCase and should contain only latin (ASCII) letters or numbers: C (diktat-
ruleset:identifier-naming)
C.kt:1:7: [IDENTIFIER_LENGTH] identifier's length is incorrect, it should be in
range of [2, 64] symbols: C (cannot be auto-corrected) (diktat-
ruleset:identifier-naming)
```

The errors, if any, are printed on the standard output.

- `plain?group_by_file`: same as above, but group errors by file, e. g.:

```
C.kt
  1:1 [MISSING_KDOC_TOP_LEVEL] all public and internal top-level classes and
functions should have Kdoc: C (cannot be auto-corrected)
  1:1 [FILE_NAME_INCORRECT] file name is incorrect - it should end with .kt
extension and be in PascalCase: C.kt
  1:1 [PACKAGE_NAME_MISSING] no package name declared in a file: C.kt
  1:7 [CLASS_NAME_INCORRECT] class/enum/interface name should be in PascalCase
and should contain only latin (ASCII) letters or numbers: C
  1:7 [IDENTIFIER_LENGTH] identifier's length is incorrect, it should be in
range of [2, 64] symbols: C (cannot be auto-corrected)
```

- `json`: report errors in the JSON format to the file named `report.json`.
- `html`: report errors in the HTML format to the file named `report.html`.

`input-paths`: custom source sets

- Default: none
- Required: **no**

One or more patterns which indicate the files or directories to check. Use a multiline string to specify multiple inputs.

- If an input is a path to a file, it is passed to *diKTat* as-is:

```
    - uses: saveourtool/benedikt@v1
      with:
        input-paths: |
          path/to/file.kt
```

- If an input is a path to a directory, the directory is recursively traversed, and all `*.kt` and `*.kts` files are passed to *diKTat*.

```
    - uses: saveourtool/benedikt@v1
      with:
```

```
          input-paths: |
            src/main/kotlin
            src/test/kotlin
```

- If an input is an *Ant*-style path pattern (such as `**/*.kt`), *diKTat* expands it into the list of files that match the path pattern. Path patterns may be negated, e. g.: `!src/**/*Test.kt` or `!src/**/generated/**`.

```
      - uses: saveourtool/benedikt@v1
        with:
          input-paths: |
            **/*.kt
            **/*.kts
            !**/generated/**
```

If this input parameter is not specified, this is equivalent to setting it to `.`, meaning *diKTat* will check all `*.kt` and `*.kts` files in the project directory unless configured otherwise.

`java-distribution` and `java-version`: running *diKTat* using a custom JVM

It's possible to run *diKTat* with a custom JVM using the `actions/setup-java` action. The following input parameters may be specified:

- `java-distribution`: the Java distribution, see the list of supported distributions.
  - Default: `temurin`
  - Required: **no**
- `java-version`: the Java version to set up. Takes a whole or semver Java version. See examples of supported syntax.
  - Default: none
  - Required: **no**

NOTE	Setting just the `java-distribution` property in order to use a custom JDK is not sufficient: you'll need to set **both** `java-distribution` **and** `java-version`:

```
      - uses: saveourtool/benedikt@v1
        with:
          java-distribution: 'temurin'
          java-version: 17
```

`fail-on-error`: suppressing lint errors

- Default: `true`

- Required: **no**

If `false`, the errors are still reported, but the step completes successfully. If `true` (the default), then lint errors reported by *diKTat* are considered fatal (i.e. the current step terminates with a failure):

```
    - uses: saveourtool/benedikt@v1
      with:
        fail-on-error: true
```

NOTE	This flag only affects the case when *diKTat* exits with code **1**. Higher exit codes are *always* fatal.

`relative-paths`: relative or absolute paths

- Default: `true`
- Required: **no**

If `true`, file paths get relativized with respect to the project directory. Otherwise, absolute file paths get reported. Example:

```
    - uses: saveourtool/benedikt@v1
      with:
        relative-paths: true
```

NOTE	When *SARIF* reporter is used, this flag has no effect: in *SARIF* mode, paths reported are always absolute.

`color`: colorizing the plain-text output

- Default: `true`
- Required: **no**

Setting this flag enables the console output to be colorized. This is only useful if the reporter is set to `plain` or `plain?group_by_file`:

```
    - uses: saveourtool/benedikt@v1
      with:
        reporter: plain
        color: true
```

`debug`: enabling debug logging

- Default: `false`

- Required: **no**

Debug logging can be enabled by setting the `debug` input parameter to `true`:

```
      - uses: saveourtool/benedikt@v1
        with:
          debug: true
```

Outputs

The action returns the exit code of the command-line client using the `exit-code` output parameter, e. g.:

```
jobs:
  diktat_check:
    name: 'diKTat Check'
    runs-on: ubuntu-latest

    steps:
      - uses: actions/checkout@v3

      - id: diktat
        uses: saveourtool/benedikt@v1

      - name: 'Read the exit code of diKTat'
        if: ${{ always() }}
        run: echo "diKTat exited with code ${{ steps.diktat.outputs.exit-code }}"
        shell: bash
```

The exit codes are documented here.

Using the command-line client

Alternatively, if you wish to run *diKTat* locally (e. g.: as a *Vim* plug-in), or you're using a different CI/CD server, you can try the command-line client.